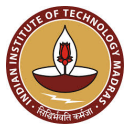
First Order Partial Differential Equations

Prof. A. K. B. Chand

IIT Madras



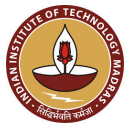
Why Partial Differential Equations?

- Many physical quantities depend on more than one variable: position and time, or two space coordinates.
- Temperature in a rod: $u(x, t)$; displacement of a membrane: $u(x, y, t)$.
- Laws of conservation and balance relate rates of change in different directions.
- Such relations lead to equations involving partial derivatives.
- Transport (advection): $u_t + c\, u_x = 0$ (first order).
- Heat conduction: $u_t = k\, u_{xx}$ (second order).
- Wave propagation: $u_{tt} = c^2 u_{xx}$ (second order).
- First order PDEs are the natural starting point: they can be reduced to ODEs.



What is a PDE?

- Let $u(x, y)$ be a function of two independent variables x and y.
- In place of ordinary derivatives we now have partial derivatives.
- First order: $u_x = \dfrac{\partial u}{\partial x}, \quad u_y = \dfrac{\partial u}{\partial y}.$
- Higher order: $u_{xx},\ u_{xy},\ u_{yx},\ u_{yy},\ u_{xxx}, \ldots.$
- An equation involving x, y, u and some partial derivatives of u is called a partial differential equation (PDE).
- u is the dependent variable; x, y are independent variables.
- We mostly deal with two or three independent variables and one dependent variable.



Notation and Order

- Throughout, $u = u(x, y)$ and we write

$$p = u_x = \frac{\partial u}{\partial x}, \qquad q = u_y = \frac{\partial u}{\partial y}.$$

- The order of a PDE is the order of the highest derivative of u in it.
- $u_x + u_y = 1$ (order 1).
- $x\,p + y\,q = u$ (order 1).
- $u_{xy} = 0$ (order 2).
- $u_{xx} + u_{yy} = 0$ (order 2).
- $u_t + u\,u_x + u_{xxx} = 0$ (order 3).
- In the beginning, we focus on first order PDEs.



Solution of a PDE

- A solution of a PDE in a domain D is a function $u(x, y)$ having the required partial derivatives in D.
- When u and its derivatives are substituted, the equation becomes an identity in D.
- Example: $u = x^2 + y^2$ satisfies $y\,u_x - x\,u_y = 0$.
- Indeed $y(2x) - x(2y) = 0$ for all (x, y).
- Also $u = \sin(x^2 + y^2)$ and $u = e^{x^2+y^2}$ satisfy the same PDE.
- In fact $u = f(x^2 + y^2)$ is a solution for every differentiable f.
- Thus a PDE generally has a huge family of solutions.
- Extra conditions (initial / boundary data) are needed to single one out.



General First Order PDE

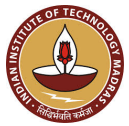
- The most general first order PDE in two independent variables is

$$F(x, y, u, p, q) = 0.$$

- Here F is a given function of its five arguments.
- With three independent variables x, y, z:

$$F(x, y, z, u, u_x, u_y, u_z) = 0.$$

- Depending on how F involves u, p, q, we get the classes.
- linear $\subset$ semilinear $\subset$ quasi-linear $\subset$ nonlinear (fully nonlinear).
- Each larger class is harder to analyse than the smaller one.

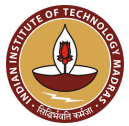


Linear First Order PDE

- F is linear if it is linear in u and its derivatives.
- The coefficients of u, p, q depend on x and y only.
- General form:

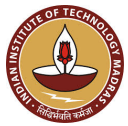
$$a(x,y)\,p + b(x,y)\,q + c(x,y)\,u = d(x,y).$$

- No products such as $u\,p$, $p\,q$, p^2 and no functions like $\sin u$.
- Equivalent description: after rationalising and clearing fractional powers, u, p, q appear only to the first degree and are not multiplied together.
- Examples: $x^2p + y^2q = u$, $\quad xy\,p + (\sin x)(\cos y)\,q = x^2y^2u$.



Homogeneous and Non-homogeneous

- In $a(x,y)p + b(x,y)q + c(x,y)u = d(x,y)$:
- If $d(x,y) \equiv 0$, the PDE is homogeneous.
- Otherwise it is non-homogeneous.
- $x\,p + y\,q - u = 0$ homogeneous.
- $p + (x+y)\,q - 5u = e^x$ non-homogeneous.
- $y\,p + xy\,u = xy$ non-homogeneous.
- $(y-z)u_x + (z-x)u_y + (x-y)u_z = 0$ homogeneous.
- For homogeneous linear PDEs, linear combinations of solutions are again solutions.



Semilinear First Order PDE

- A PDE that is not linear is called nonlinear.
- Some nonlinear classes are still easy to handle.
- A first order PDE is semilinear if the coefficients of p and q depend on x, y only.
- The remaining term may be nonlinear in u.
- General form:

$$a(x,y)\,p + b(x,y)\,q = c(x,y,u).$$

- Example: $p + q = u^2$.
- Example: $x\,p + y\,q = \sin u$.
- Every linear PDE is semilinear (take $c = d - c_0 u$).



Quasi-linear First Order PDE

- A first order PDE is quasi-linear if F is linear in p and q.
- The coefficients of p, q may now involve u as well.
- General form:

$$a(x, y, u)\, p + b(x, y, u)\, q = c(x, y, u).$$

- Example: $x(y^2 + u)\, p - y(x^2 + u)\, q = (x^2 - y^2)\, u$.
- Example: $u\, p + q + u^2 = 0$ (inviscid Burgers type).
- Example: $(y^2 - u^2)\, p - xy\, q = xu$.
- This equation is also called Lagrange's linear equation.
- We will study the solution of Lagrange's linear equation - Main object of study in the first part.



Fully Nonlinear First Order PDE

- If F is not linear in p and q, the PDE is fully nonlinear.
- Example: $p^2 + q^2 = 1$ (eikonal equation).
- Example: $p\,q = u$.
- Example: $u = p\,x + q\,y + p\,q$ (Clairaut type).
- No single general method like Lagrange's exists for them.
- Hierarchy:
- linear $\Rightarrow$ semilinear $\Rightarrow$ quasi-linear, but not conversely.
- Always identify the class before choosing a method.



Classify the Following

- $x\,p + y\,q = u$ — linear, homogeneous.
- $p + (x + y)\,q = 5u + e^x$ — linear, non-homogeneous.
- $p + q = u^2$ — semilinear.
- $u\,p + q = 0$ — quasi-linear.
- $x(y^2 + u)\,p - y(x^2 + u)\,q = (x^2 - y^2)u$ — quasi-linear.
- $p^2 + q^2 = u^2$ — nonlinear.
- $p\,q = 4xyu$ — nonlinear.
- Check: does u appear in the coefficient of p or q? Is F linear in p, q?



Solving by Direct Integration

- Sometimes ODE techniques solve a PDE directly.
- This works when derivatives can be integrated w.r.t. one variable.
- Or when a substitution such as $v = u_x$ produces an ODE.
- Integrating w.r.t. x treats y as a constant.
- Hence the "constant" of integration may be any function of y.
- So, arbitrary constants of ODEs become arbitrary functions.
- Typically the general solution of an nth order PDE contains n arbitrary functions.
- Initial and boundary conditions are used to determine them.



Example 1: $u_x = x + y$

- Keep y fixed and integrate w.r.t. x

$$u(x, y) = \int (x + y)\, dx = \frac{x^2}{2} + xy + f(y).$$

- The integration constant cannot depend on x, but may depend on y.
- So we write it as $f(y)$, an arbitrary function of y.
- Check: $u_x = x + y + 0$ ✓.
- One first order PDE $\Rightarrow$ one arbitrary function.
- E.g. $u = \frac{x^2}{2} + xy + \cos y$ is a particular solution.



Example 2: $u_{xy} = 0$

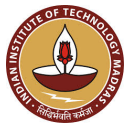
- Integrate w.r.t. y (keeping x fixed): $u_x(x, y) = f(x)$.
- Integrate w.r.t. x (keeping y fixed):

$$u(x, y) = \int f(x)\, dx + g(y).$$

- Since f is arbitrary, its antiderivative $h(x)$ is also arbitrary.
- General solution:

$$u(x, y) = h(x) + g(y).$$

- h and g are arbitrary (differentiable) functions.
- Second order PDE $\Rightarrow$ two arbitrary functions.
- Initial and boundary conditions are imposed after finding the general solution.



Example 3: $u_{xx} = y^2 \cos 2x$ with Boundary Data

- Find $u(x, y)$ with $u(0, y) = 0 = u(\pi/2, y)$.
- Integrate once w.r.t. x:

$$u_x = y^2 \frac{\sin 2x}{2} + f(y).$$

- Integrate again w.r.t. x:

$$u = -y^2 \frac{\cos 2x}{4} + x f(y) + g(y).$$

- Two arbitrary functions $f(y)$ and $g(y)$ appear.
- They are fixed using the two boundary conditions.

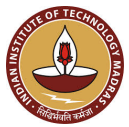


Example 3 (contd.)

- $u(0,y) = 0 \Rightarrow -\dfrac{y^2}{4} + g(y) = 0 \Rightarrow g(y) = \dfrac{y^2}{4}.$
- $u(\pi/2, y) = 0 \Rightarrow \dfrac{y^2}{4} + \dfrac{\pi}{2} f(y) + \dfrac{y^2}{4} = 0.$
- Hence $f(y) = -\dfrac{y^2}{\pi}.$
- Solution:

$$u(x,y) = \frac{y^2}{4}(1 - \cos 2x) - \frac{xy^2}{\pi} = y^2\left(\frac{\sin^2 x}{2} - \frac{x}{\pi}\right).$$

- Check: $u_{xx} = y^2 \cos 2x$, $u(0,y) = 0$, $u(\pi/2, y) = y^2(\frac{1}{2} - \frac{1}{2}) = 0$ ✓.



Practice: Direct Integration

- Solve $u_y = 2xy + \cos x$.
- Solve $u_{xx} = 6x$ with $u(0, y) = y$, $u_x(0, y) = y^2$.
- Solve $u_{xy} = x + y$.
- Solve $u_{yy} = \sin y$ with $u(x, 0) = x$, $u(x, \pi) = 0$.
- Solve $u_x = u$ (treat as ODE in x): answer $u = f(y)e^x$.
- Solve $u_{xy} = u_x$ (put $v = u_x$).
- In each case count the arbitrary functions in the general solution.
- Verify each answer by substitution.

